1. Clinical Question & Core Hypothesis

Clinical Question

Why do some tumors with abundant immune infiltrates nonetheless recur? The path suggests a link between SLUG-driven EMT, LZTS2-associated prognostic signaling, and recurrence, intersecting with paracrine PDGFC and mTOR network activity and age-related immune decline. 2) Tier 1 paths used: I built on the LZTS2 → recurrence-free survival link (prognostic anchor) and the SLUG ↔ PDGFC EMT/regulatory loop (mechanistic anchor) because these directly explain how tumor-intrinsic programs could negate immune infiltration. 3) Tier 2/supporting evidence: PDGFC and mTOR signaling (supporting nodes) provide plausible pathways by which SLUG activity could remodel tumor behavior and promote immune evasion; age-related CD8+ T cell decline explains host-context modulation of immune control. 4) Endpoint choice: PFI (progression-free interval) maps most closely to recurrence-free survival in the path and is robustly available in TCGA-BRCA for testing immune–recurrence relationships. 5) Deliberate exclusions: I excluded broader lists of proliferation/metabolic regulators and clinical-surgical notes (AUR/BPH) to keep the hypothesis sharply mechanistic and testable within TCGA-BRCA imaging and molecular data; those factors are left as exploratory variables in secondary_modifiers?

Working Hypothesis

Primary question: In breast cancers with high Tumor-Infiltrating Lymphocyte (TIL) density, does concurrent SLUG-driven EMT coupled with LZTS2 suppression create an immune-evasive state that diminishes the favorable prognostic impact of TILs, leading to earlier progression (shorter PFI)? Secondary modifiers and predictions: (1) This negative effect of high TIL density on PFI will be amplified when SLUG is high and LZTS2 expression is low. (2) Elevated PDGFC signaling and MTOR pathway activity will strengthen the SLUG/LZTS2-driven immune-evasive state, further shortening PFI. (3) Reduced CD8A expression (a proxy for diminished cytotoxic T cell function) will worsen the TIL-associated risk, particularly in older patients where immune competence declines. (4) Age will modulate the magnitude of these interactions, with stronger effects in older individuals. In TCGA-BRCA, we predict: among tumors with high TIL density, those with SLUG-high and LZTS2-low will show shorter PFI, and this association will be most pronounced in older patients and in samples with high PDGFC/MTOR activity and low CD8A expression.

Element	Value
Imaging feature	Tumor-Infiltrating Lymphocyte (TIL) density
Anchor gene(s)	LZTS2, SLUG
Primary endpoint	PFI

2. Why This Gene — Not Others

Primary hypothesis gene(s)

LZTS2: LZTS2 → recurrence-free survival link (prognostic anchor) and the SLUG ↔ PDGFC EMT/regulatory loop (mechanistic anchor) because these directly explain how tumor-intrinsic programs could negate immune infiltration.

SLUG: SLUG-driven EMT, LZTS2-associated prognostic signaling, and recurrence, intersecting with paracrine PDGFC and mTOR network activity and age-related immune decline.

Alternative / secondary genes considered

Tier	Gene	Role / Reason excluded from core
Secondary	PDGFC	Supporting modifier from knowledge graph
Secondary	MTOR	Supporting modifier from knowledge graph
Secondary	CD8+ T cell	Supporting modifier from knowledge graph
Secondary	age	Supporting modifier from knowledge graph

3. Clinically Testable Predictions

Prediction 1: drives EMT and is implicated in immune evasion; LZTS2 functions as a tumor suppressor whose loss can alter cell polarity and signaling relevant to recurrence risk; PDGFC can activate PDGFR and downstream mTOR signaling, modulating tumor–stroma interactions and potentially dampening anti-tumor immunity; CD8A reflects cytotoxic T cell presence/function, which declines with age and can be functionally impaired in tumors with EMT programs.

Prediction 2: , LZTS2, PDGFC, MTOR pathway genes, and CD8A, plus PFI endpoints, enabling interaction-focused analyses. Experimental testing will rely on regression models with interaction terms, stratified analyses by age, and mediation/moderation assessments to isolate the proposed axis while controlling for confounders.

Prediction 3: In retrospective cohort data, the prognostic benefit of the imaging feature will be modified by the anchor gene expression level (significant interaction term).

4. Minimum Viable Validation Plan

A. Retrospective Cohort Analysis

Data source: TCGA-BLCA (Firehose Legacy), ~400 patients - Cox PH model with interaction term for the imaging feature and anchor gene - Kaplan-Meier 2x2 stratification

B. Tissue-Level Validation

Methods: Multiplex IF / IHC / spatial transcriptomics

Key readouts: - Immune structure markers (e.g., CD20, CD3, PNAd) - Effector cell localization (CD8) relative to tumor nests - Anchor gene protein / phospho-protein expression - Stromal features (collagen density by SHG, if applicable)

Goal: Confirm that the proposed mechanism operates at the spatial / tissue level.

C. Functional Validation

Models: Ex vivo organotypic cultures, 3D co-culture, syngeneic mouse models

Perturbations: Genetic (KO/rescue) or pharmacologic modulation of anchor gene

Readouts: T-cell infiltration, cytotoxicity (granzyme B / IFN-gamma), tumor growth / survival

Goal: Establish causal relationship between anchor gene activity and immune efficacy.

5. Strengths, Caveats & Alternative Directions

Strengths

- Interaction-first biomarker strategy — asks WHY the imaging feature sometimes fails, not just whether it predicts outcome.

Caveats & Risks

- Hypothesis is computationally derived and requires experimental validation.

Alternative Directions

- PDGFC: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.
- MTOR: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.